

Rowan Torbitzky-Lane

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Education

B.S. Computer Science - Minor in Applied Mathematics - GPA: 3.80

December 2024

B.S. Mathematics - GPA: 4.00

May 2027

Skills

- | | | |
|----------|---------|-----------|
| • Python | • Godot | • CUDA |
| • Rust | • SQL | • Haskell |
| • Docker | • Linux | • Nix |

Relevant Coursework

- | | | |
|----------------------|-----------------------------|-----------------------------|
| • Algorithms | • Differential Equations | • Linear Algebra I/II |
| • Data Structures | • Calculus III | • Physics I/II |
| • Introduction to AI | • Prob and Stat (MTH 34100) | • Linear Regression (@ MST) |

Experience

Computer Vision Intern - USC Institute for Creative Technologies

May 2026 - August 2026

- Train Gaussian Splats on bounded and unbounded scenes
- Extract single objects from Gaussian Splats
- Generate Meshes from Gaussian Splats
- Utilize Nix for programming environment setup

NSF Research Assistant - Missouri University of Science and Technology

May 2024 - October 2024

- Train novel, bi-directional LLMs on Genome Sequences to correct errors in reads
- Clean and analyze textual data with Scikit-learn and Scikit-bio
- Incorporate huggingface library for model testing

Research Assistant - Center for Intelligent Infrastructure

May 2023 - October 2023

- Calculate Optical Flow with MATLAB and C++ utilizing Slurm on campus HPC
- Coordinate Web Requests with Django to interact with internal PostgreSQL server
- Improve frontend with Typescript, such as adding video upload capability
- Extract image features using OpenCV such as utilizing optical flow to predict annotations in subsequent video frames
- Perform Annotations for image segmentation Convolutional Neural Networks

Data Science Co-Op - Hunter Engineering Company

January 2022 - August 2022

- Utilize YOLO and PyTorch in license plate recognition problems
- Pioneer object detection models to eradicate bad data before annotators see it
- Innovate data pipeline by automating our data downloading process
- Design and containerize APIs with FastAPI and Flask within Docker, used to create endpoints within Azure
- Leverage Pandas and Microsoft SQL in data processing pipelines
- Implement syntax formatting actions into internal Azure DevOps git repository ensuring up to par formatting
- Utilize Excel for data annotation involving writing macros to automate data exportation
- Transition to storing data in AWS S3 Bucket Storage

Personal Projects

Self-Hosted NixOS Server - SQL, YOLO, Docker, Firewall, Git, Nginx, CUDA

January 2023 - Present

- Train bird object detection Convolutional Neural Networks
- Host game servers via Docker
- Point domain names via A records

Rust Genetic Programming Framework - Polars, rust_decimal

March 2025 - Present

- Perform pre-processing on Time Series Data
- Utilize test driven development philosophy involving writing unit tests
- Write declarative and procedural macros for reduced boilerplate
- Utilize rust_decimal's floating point precision for increased accuracy

Invited Talks

Genetic Programming in Python @ PySTL https://www.youtube.com/watch?v=ZzYi_j2IYuY

September 18th, 2025

Actix Web Framework @ STL Rust <https://rumble.com/v72mid6-actix-web-unleashed.html>

December 4th, 2025